

Unit

3

Lesson

2



Beyond Imagination.

Tap to Help.

Time Frame: [90 min]

Lesson Objectives:

By the end of this lesson students will be able to:

- Understand how components work in mobile apps
- Arrange and label UI (user interface) components to help users navigate your app
- Apply simple UX (user experience) ideas to make your design more helpful
- Understand how components work in mobile apps

Challenge questions of the lesson

- How can the way an app looks affect the way someone uses it?
- Why might a clear layout be more important in a health app than in a game app?
- Which components help make an app easier for someone with poor memory or vision?
- What would you change in your app design if your user was 75 years old and not used to phones?
- Can a good design keep someone safe? How?

SEL Relation

- **Empathy:** Students imagine themselves in someone else's shoes—perhaps an elderly person who forgets to take medicine, someone with vision loss, or someone who struggles to use small touchscreens.
- **They begin to ask:** “What would they need from my app? What would make them feel confident, supported, and safe?”
- **Responsibility:** When students realize their app could truly impact someone's daily life, it becomes more than just a school project. They start making intentional design choices, like using bigger fonts, voice features, or clear color contrasts—not because they were told to, but because they see the value of inclusive design.
- **Creative Problem-Solving:** Students are challenged to solve a specific user-centered problem—How can this app actually help someone?
- This activates creative thinking in a meaningful context, where aesthetics must support usability, not just style.

Engage



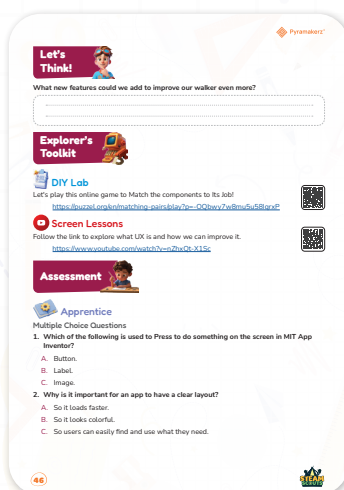
Teacher's Role:

- Begin the lesson by revisiting the walker students designed or explored in the previous activity. Prompt them to think about how engineers improve designs over time.
- Display the prompt from the student book:
- “What new features could we add to improve our walker even more?”
- Guide students through a short discussion that activates empathy and innovation. Encourage them to think from the perspective of someone elderly or with mobility challenges.
- Use open-ended questions to spark ideas:
 - “What might make the walker easier to use at night or outdoors?”
 - “What could help if the user is carrying things or walking for a long time?”
 - “What would you add to make the walker feel smarter or more helpful?”
- Reinforce that this is a creative brainstorming moment. There are no right or wrong answers—every idea is valuable.

Expected from Students:

- Reflect on their previous walker design and its current limitations.
- Brainstorm 2–3 new feature ideas to improve the walker’s safety, comfort, or usability.
- Write their suggestions clearly in the space provided.
- Be prepared to share their ideas in pairs or with the full class.
- Use imagination, empathy, and prior knowledge of assistive tools.

Parts of the book included.



Let's Think!



Objective:

Students will apply design thinking and user-centered empathy to critically evaluate their current walker prototype and propose thoughtful improvements. The goal is to encourage iterative design by analyzing functionality, user needs, and real-world application in assistive technologies.

Answers:

Question: What new features could we add to improve our walker even more?

Sample Answers:

- We could add LED headlights
 - This would help users walk safely in the dark or in low-light areas.
- We could install an emergency SOS button
 - If the user falls or needs help, they can press the button to alert someone immediately.
- We could add a GPS tracking system
 - This would help caregivers locate the user in case they get lost.
- We could design a foldable seat into the walker
 - It allows users to take a rest during long walks.
- We could attach a small storage basket
 - This would let users carry items like water bottles, phones, or medication without needing an extra bag.
- We could add voice-command features
 - The user could tell the walker to stop or turn on the lights, making it easier for those with limited hand mobility.

Tip: Encourage students to justify their ideas not just with "coolness," but based on user challenges, safety, and efficiency. Ask: "Who will benefit from this feature, and how?"

Possible Strategies for Implementation:

1. Think-Pair-Share

Begin with individual brainstorming time where students write their answers to the question. Then, they pair up to compare and refine their ideas. Finally, selected pairs share with the whole class. This strategy supports confidence-building and encourages collaborative thinking.

2. Empathy-Driven Reflection

Ask students to imagine themselves as elderly individuals using a walker. Use prompts like:

- "What daily challenges might I face?"

- “How could a walker help me feel safer or more independent?”
- This helps shift thinking from function to real-world user experience.

3. Design Wall or Digital Idea Board

Have students post their best ideas on a class wall (sticky notes) or digital board (Padlet, Jamboard). Group similar ideas into categories (e.g., safety, comfort, mobility) and invite students to comment or vote on the most impactful ones.

4. Sketch and Annotate

After writing their ideas, ask students to choose one feature and sketch how it would look on the walker. They should label its parts and describe how it works. This connects ideation with visual design and real-world application.

5. Use of Real-Life Examples for Inspiration

Show brief case studies, images, or short videos of real smart devices and assistive tools. Then ask:

- “What do you notice?”
- “How are these designs helpful?”
- “What can we learn from these examples to improve our walker?”

6. Structured Sentence Starters

Provide sentence scaffolds for students who may need language support:

- “I would improve the walker by adding ____.”
- “This feature would help the user by ____.”
- “If someone was ____, then the walker could ____.”

Explorer's Toolkit



Teacher's Role:

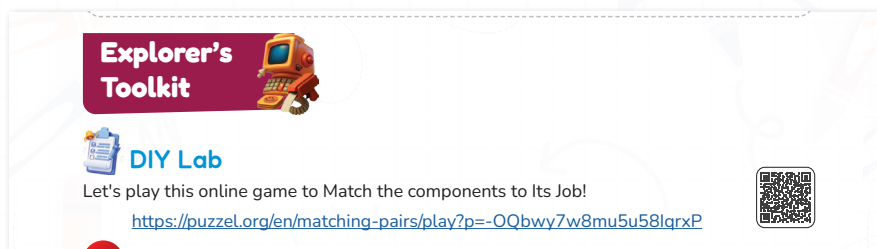
- Introduce the activity by explaining that understanding the function of each component is essential before students can begin building or improving any system.
- Provide context for the activity by linking it to the walker project: “Before we can build or improve a smart walker, we need to know how each electronic or mechanical part works. This matching game will help you make those connections.”
- Monitor students as they complete the game individually or in pairs, offering clarification as needed on each component’s function.
- Encourage students to discuss their thinking and justify their answers if working collaboratively.
- After the activity, lead a short debrief discussion using questions like:
 1. “Which component surprised you the most?”
 2. “How might this part help us improve our walker design?”

3. “Were there any pairs you struggled to match? Why?”

Expected from Students:

- Engage with the online matching game by pairing each component with its correct function.
- Demonstrate understanding of the roles of basic parts (e.g., sensors, motors, buttons, LEDs) through active participation in the game.
- Apply what they’ve learned by reflecting on how each component could be used in the walker project.
- Collaborate respectfully if working in pairs, discussing choices and reasoning together.
- Participate in the class debrief to share insights and questions that emerged during the activity.

Parts of the book included.



DIY Lab

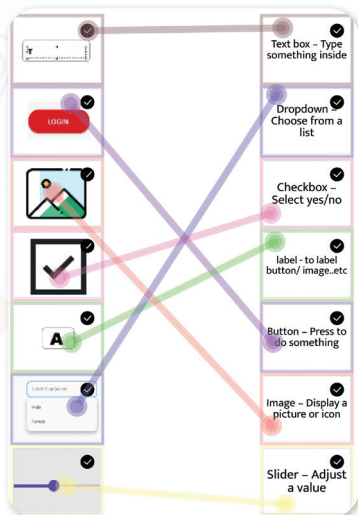
Objective:

Students will identify and understand the functions of various electronic and mechanical components used in assistive technologies. Through a digital matching activity, students will reinforce their knowledge of how each part contributes to a larger system. This foundational understanding will prepare them to make informed decisions when building or improving their smart walker prototype.

Resource: <https://puzzel.org/en/matching-pairs/play?p=-OQbwy7w8mu5u58lqrxP>



Model answer:



Component	Function
Button	Button – Press to do something
Label	Label – to label button/ image
Text Box	Text box – Type something inside
Checkbox	Checkbox – Select yes/no
Slider	Slider – Adjust a value
Image	Image – Display a picture or icon
Dropdown	Dropdown – Choose from a list

Activity instructions:

Step 1: Set the Stage

Implementation:

- Say to students: “Every app you’ve ever used is built with tiny tools called components. A component can be a button, a text field, or a checkbox. But here’s the catch—not all components are created equally. Some are helpful. Others can confuse people.”
- Then Ask: “Have you ever seen an app screen and didn’t know where to click or what to do next? What if that screen was supposed to remind someone to take their medicine?”
- Why it matters: This ties the activity to SDG 3: Good Health and Well-Being and develops empathy for users who need support.

Step 2: Distribute Game Materials

Option A: Digital

- Direct students to the online version or Google Slides version of the game
- Pair them up with a partner to play and discuss

Option B: Printable

- Give each group:
 - 1 set of Component Cards
 - 2–3 App UI Screenshot pages
- Have scissors or glue ready if they need to assemble their cards

UDL Tip: Let students choose to work solo, in pairs, or triads.

Note: flexible grouping gives student choice and builds confidence

Step 3: Matching Game

Instructions to students: “Your job is to match each component card to the part of the app screenshot where that component is being used.

For example, a button might be the thing that says ‘Remind Me’ or ‘Save’.

- Ask these discussion questions while circulating:
 - “What is this component helping the user do?”
 - “Would this be easy or hard for someone to use every day?”
 - “What would happen if the app didn’t have this component?”
- UDL Tip:
 - Some students may need a visual cheat sheet with examples
 - Others may prefer to write their matches rather than use cards

SEL Integration (through teacher tone): “Remember, we’re not just playing a matching game—we’re designing for real people. What if your grandmother had to use this app—would she know what to press?”

Step 4: Group Reflection & Share-Out

Ask each group to share 1 component they found important and explain why.

Optional student prompts: “The button is important because it’s how the user tells the app what to do.”

“The checkbox helps people keep track if they took their medicine.”

SDG Reminder:

Remind students this kind of thinking is part of real-world problem solving. They’re learning how to design for well-being.

Possible Strategies for Implementation:

1. Pre-Activity Concept Review

Begin with a quick, visual review of the components students will encounter in the game (e.g., LED, buzzer, motor, switch, sensor). Use physical examples or a slide presentation to introduce each part and its function before students begin the activity.

2. Guided Exploration in Pairs

Pair students to encourage discussion and collaborative problem-solving as they match each component to its job. Prompt them to explain their reasoning to each other before confirming a match. This supports peer learning and verbal processing of technical vocabulary.

3. Use of Component Reference Sheet

Provide a printed or digital component glossary or diagram students can reference during the activity. This scaffolds the learning process and supports students who may be unfamiliar with some parts.

4. Whole-Class Debrief and Reflection

After completing the game, conduct a class discussion using prompts such as:

- “Which component was most challenging to identify? Why?”
- “Which part do you think is most important for a walker? Why?”

This reflection helps deepen understanding and link the virtual activity to the physical design context.

5. Anchor the Activity in the Real-World Project

Ask students to select one component from the game and explain how it could be used to improve the smart walker. For example:

- “How would adding a buzzer help an elderly user?”
- “Where might a light sensor be placed on the walker, and why?”

6. Use of Digital Progress Tracking

Have students take a screenshot or submit a reflection form after completing the game to document their participation and learning outcomes. This can serve as a formative assessment tool.

Explain



Teacher's Role:

- Set a purpose for watching the video through real-world connections
- Scaffold understanding of new vocabulary like “component,” “layout,” “arrangement,” etc.
- Encourage active observation and student-led discovery
- Create an inclusive space for students to ask questions, make predictions, and reflect
- Help students connect the video to their own app design idea

Expected From Students:

- Watch attentively and take note of at least 3 key components (e.g., button, label, textbox)
- Reflect on how each component helps the user
- Think critically about how they would use these components in their own app design
- Participate in post-video discussion or written reflection

Parts of the book included.

<https://puzzet.org/en/matching-pairs/play?p=-UQbwy/w8mu5u58iqrxP>



Screen Lessons

Follow the link to explore what UX is and how we can improve it.

<https://www.youtube.com/watch?v=nZhXQt-X1Sc>



Screen Lessons

Objective:

Students will understand the meaning of User Experience (UX) and how it applies to product design. By exploring real-life examples and concepts from the video, they will learn how thoughtful design decisions can improve the experience, comfort, and safety of users—especially when developing assistive technologies such as a smart walker.

Video Title: What is UX Design?

Video Link: <https://www.youtube.com/watch?v=nZhXQt-X1Sc>



Possible Strategies for Implementation:

1. Before Watching (Pre-Watching)

- **Activate Prior Knowledge:**
 - Ask: “Have you ever used something that was hard to use or confusing?” Let students briefly share examples (e.g., tricky apps, uncomfortable chairs, devices with too many buttons).

- **Set a Purpose for Viewing:** Write the viewing question on the board:
 - “What is UX, and why is it important for designing something like a smart walker?”
- **Introduce Key Vocabulary:** Preview terms such as:
 - UX (User Experience)
 - Accessibility
 - Design Thinking
 - Interaction

2. During Watching (Active Engagement)

- **Focused Viewing Task:** Ask students to write down:
 - One definition or description of UX.
 - Two examples of how design affects the user’s experience.
 - One way they could apply UX to improve their walker.
- **Pause & Discuss Points:**
 - Stop at key timestamps to check understanding or highlight a concept (e.g., when the narrator explains how a simple door handle affects UX).
- **Guided Notes Template (optional):**
 - Provide a worksheet with prompts or sentence starters (e.g., “UX matters because...”, “Good UX helps users by...”)

3. After Watching (Post-Watching)

- **Whole-Class Discussion:** Discuss:
 - “What surprised you about UX?”
 - “Why is it important to think about UX when designing for someone elderly or with mobility challenges?”
- **Connect to the Project:**
 - Ask: “How can we improve the UX of our walker design? What features or layout changes would make it more user-friendly?”
- **Reflection Prompt (Individual or Group):** Have students complete the sentence:
 - “One UX idea I will apply to my walker is...”
 - Then share or post on a class design wall.
- **Extension (Optional):**
 - Let students explore poor UX examples in everyday life (e.g., confusing signs, awkward interfaces) and redesign them as a short challenge.

Elaborate



Teacher's Role:

- Facilitate, not grade — this is a low-stakes check to help students and you know what they've learned
- Encourage confidence, not perfection
- Offer multiple ways to show understanding (reading, drawing, choosing)
- Support students who need scaffolding (read-aloud, visual aids, translation)

Tone to use: "Let's see how much you've discovered so far. Don't worry about getting it all right. This will help us plan your next steps as app designers."

Expected From Students:

- Answer a short set of objective questions (multiple choice + fill-in-the-blank)
- Use what they learned from the DO IT, WATCH IT, and READ IT parts
- Apply terms like component, Designer, Blocks, UX, and layout
- Think about why clear design matters, not just what tools are used

Parts of the book included.



Apprentice

Multiple Choice Questions

1. Which of the following is used to Press to do something on the screen in MIT App Inventor?

- A. Button.
- B. Label.
- C. Image.

2. Why is it important for an app to have a clear layout?

- A. So it loads faster.
- B. So it looks colorful.
- C. So users can easily find and use what they need.

3. What is the main job of the Blocks Editor in MIT App Inventor?

- A. To decorate the app.
- B. To code how the app behaves.
- C. To upload the app to Google Play.

Fill-in-the-Blank

1. A is a building block in App Inventor like a button, label or textbox that is added to the app screen.

2. The screen where you design the look of your app is called the screen.



Apprentice

Objective:

Students will demonstrate their understanding of core App Inventor interface elements and user experience (UX) principles by answering structured multiple choice and fill-in-the-blank questions. This assessment checks their ability to recall key terminology, describe the function of screen components, and explain the importance of clear layout and design logic in digital applications.

Answers:

Assessment



Apprentice

Multiple Choice Questions

1. Which of the following is used to Press to do something on the screen in MIT App Inventor?
☐ A. Button.
☐ B. Label.
☐ C. Image.
2. Why is it important for an app to have a clear layout?
☐ A. So it loads faster.
☐ B. So it looks colorful.
☐ C. So users can easily find and use what they need.

3. What is the main job of the Blocks Editor in MIT App Inventor?

- ☐ A. To decorate the app.
- ☐ B. To code how the app behaves.
- ☐ C. To upload the app to Google Play.

Fill-in-the-Blank

1. A component is a building block in App Inventor like a button, label or textbox that is added to the app screen.
2. The screen where you design the look of your app is called the Designer screen.

Possible Strategies for Implementation:

Strategy 1: “Choice of Voice” Assessment Format

UDL-aligned

How to implement:

- Offer students multiple ways to complete the FOCUS questions:
 - On paper
 - Verbally (teacher asks, student answers aloud or in pairs)
 - Google Form or digital tool (e.g. Quizziz or Google Slides)
 - Drawing: sketch and label a screen layout instead of filling blanks

Why this helps: Supports learners with reading or writing challenges, and increases engagement.

Strategy 2: Think-Aloud Partner Check

How to implement:

- Students complete the first 2–3 questions independently
- Then turn to a partner and discuss their reasoning:
 - “Why did you choose this component?”

- “What made you think that’s what the Blocks Editor is for?”

Why this help: Builds confidence, strengthens communication, and supports students who benefit from verbal reasoning.

Strategy 3: Confidence Scale Check-In

SEL Strategy: Self-awareness + Mindset

How to implement:

- After completing the 5 questions, ask students:
 - “On a scale from 1 to 5, how confident do you feel about building your app screen now?”
 - Let them color a number, use thumbs-up emojis, or move a sticky note.
 - Follow up with: “What’s one thing you want to learn more about?”

Why this helps: Encourages reflection, self-monitoring, and normalizes that learning is a process.

Elaborate



Teacher's Role:

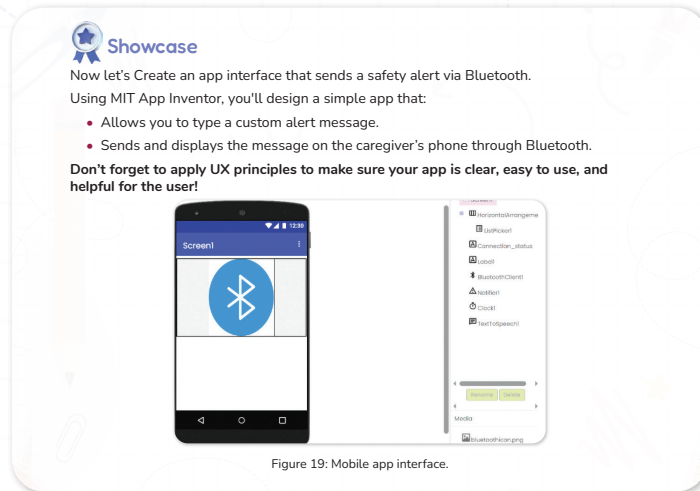
- Coach: Guide students to clarify their thinking and design choices
- Problem-Solving Facilitator: Support technical fixes and debug moments
- Empathy Encourager: Help students explain how their app helps people
- Inclusion Leader: Ensure all students can share or show their work in a way that fits their learning style
 - Say: “Show me how your app helps someone feel safe.”
 - Say: “If someone with a disability were using your app, what would they notice first?”

Expected From Students:

- **Build a working prototype of a Smart Alert App**
- **Include:**
 - A TextBox to write the alert
 - A Button to send it
 - BluetoothClient and BluetoothServer components
 - A Notifier or Label to display the alert
- **Demonstrate how their app:**
 - Sends an alert from one device (elder)
 - Receives it on another (caregiver)

- **Explain:**
 - Why they chose their design
 - How it helps someone stay safe
 - How they tested and improved it

Parts of the book included.



Showcase

Objective:

Students will design and present a functional mobile app interface in MIT App Inventor that sends a custom safety alert via Bluetooth. Through this challenge, students will apply their understanding of user experience (UX) design principles to create an app that is not only technically functional but also clear, accessible, and user-friendly. This task demonstrates their ability to integrate coding, interface design, and empathy-driven thinking in a real-world assistive technology scenario.

Recourses link: https://drive.google.com/file/d/1SkQHt5h6mXWYMqErXL8UtcLIYmGmsUrD/view?usp=drive_link



Reflection Questions (with sample student answers):

1. How does your app help someone feel safe or supported?

"It helps someone who might fall or get stuck send a quick alert to their caregiver."

2. What's one thing you would improve or add to your app?

"I'd add sound or a vibration when the alert is received, so it doesn't get missed."

3. What did you learn about Bluetooth or real-time messaging?

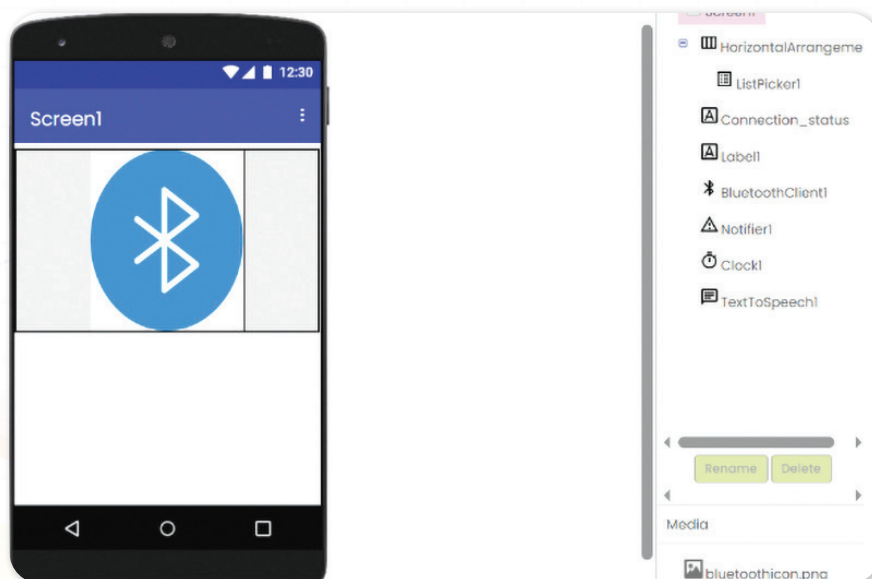
"I learned that you don't need the internet to send messages — Bluetooth connects devices directly."

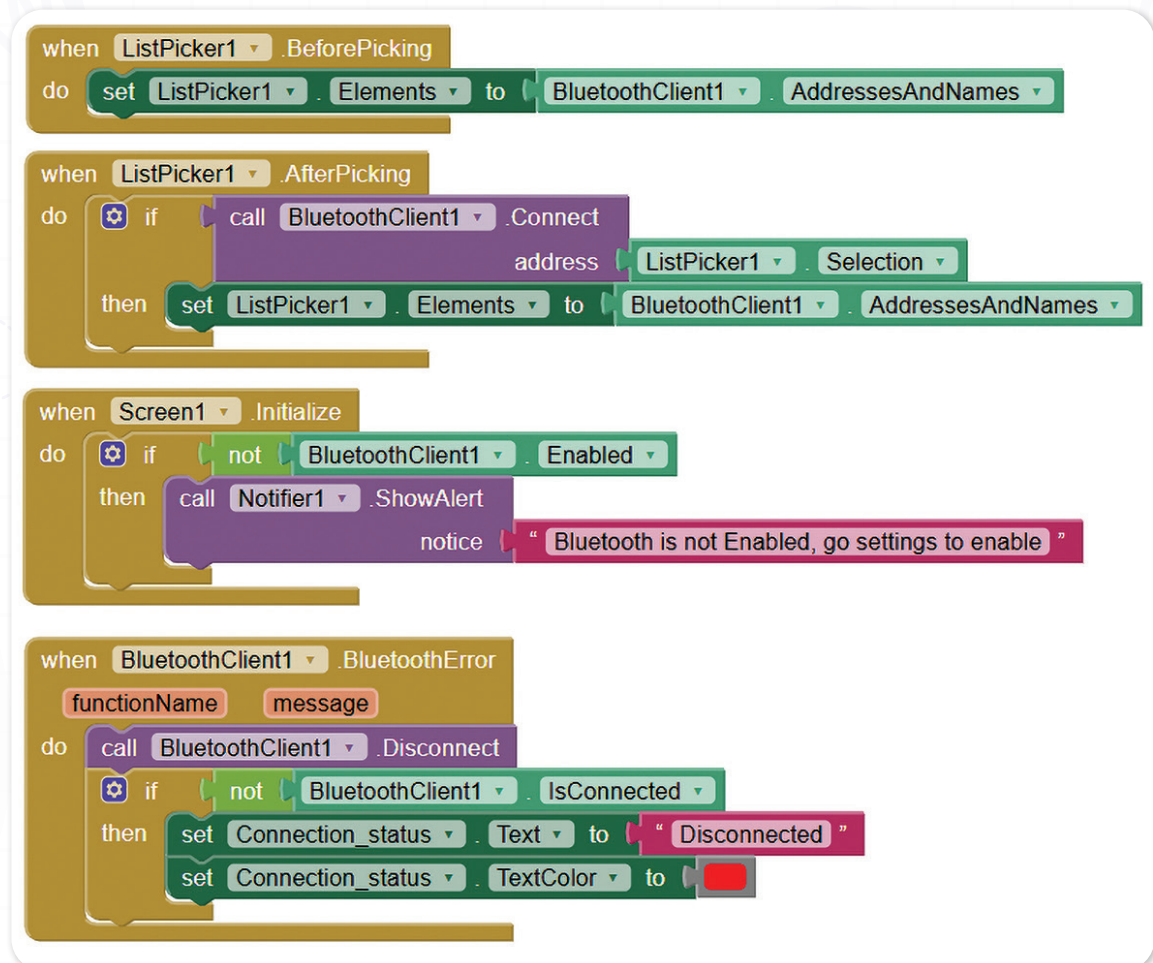
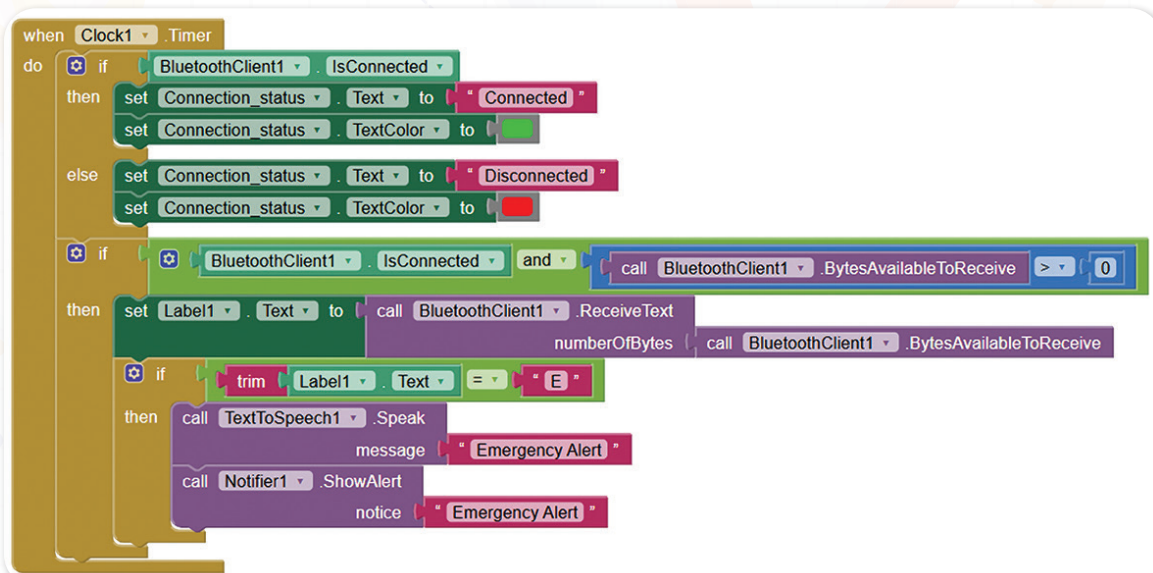
4. What made you feel proud during this activity?

When my partner saw the alert pop up on their phone — it felt like we built something real that could help someone.

Step-by-Step Instructions:

- **Open MIT App Inventor and start a new project:**
 - SmartAlertShowcase
- **In the Designer view, add:**
 - Label: Title (e.g., “Smart Alert System”)
 - TextBox: Write the alert message
 - Button: “Send Alert”
 - ListPicker: Choose device to send to
 - Label/Notifier: Display received alert
 - BluetoothClient & BluetoothServer: Non-visible
- **In the Blocks view, program:**
 - When ListPicker.BeforePicking → list available devices
 - When ListPicker.AfterPicking → connect to device
 - When Button.Click → send the alert message
 - When BluetoothServer.MessageReceived → display the message using Notifier or Label
- **Test your app in pairs:**
 - One student = elder device (sends message)
 - One student = caregiver device (receives message)
- **Present or demonstrate:**
 - What your app does
 - Who it helps
 - What you learned while testing and fixing it





Possible Strategies for Implementation:

1. Clarify the Design Expectations Upfront

Before students begin, revisit the UX principles introduced during the Explain phase:

- Clear layout
- Logical navigation
- Accessibility and ease of use

Ask students to define what makes an app "user-friendly" and write their own UX checklist to guide their design.

2. Break Down the Task into Milestones

Divide the app creation process into smaller, achievable parts:

- Milestone 1: Add a text box for typing a message
- Milestone 2: Add a "Send" button with a Bluetooth component
- Milestone 3: Program the Blocks Editor to send the message to another device
- Milestone 4: Test the interface for clarity and ease of use

Use a visible class checklist or progress tracker so students can self-monitor and stay on task.

3. Encourage UX-Driven Peer Feedback

Have students work in pairs or small groups to test each other's app layout before coding is finalized:

- Use a structured peer feedback form with guiding questions:
 - "Was it clear where to type?"
 - "Is the message easy to send?"
 - "Was anything confusing about the interface?"

This helps students apply real UX critique in a meaningful way.

4. Provide Code Snippets and Templates

To support different skill levels, provide optional starter blocks or coding templates for the Bluetooth connection and message transfer. Allow more advanced students to customize or extend functionality.

5. Use the EDP: Emphasize Iterate and Improve

Reinforce that iteration is a key part of the engineering design process. After the first round of testing, ask:

- "What could you simplify?"
- "Where might someone get confused?"
- "Can a visual element (like color or icon) improve usability?"

Let students make changes based on their reflection and peer feedback.

6. Showcase with Purpose

Conclude with a mini "App Pitch" session where students:

- Present their app to the class or another group
- Explain what the app does
- Describe how they applied UX principles
- Highlight how the app supports user needs (e.g., elderly or caregiver)

This reinforces communication skills and ensures students are thinking beyond just functionality.

7. Reflection and Self-Evaluation

Have students complete a self-evaluation form:

- “What feature are you most proud of?”
- “How did you apply UX principles?”
- “What would you improve if you had more time?”

Rubric – Smart Alert App Prototype (10 Points Total):

Criteria	Excellent (2 pts)	Good (1 pt)	Needs Work (0 pts)
Interface Design	Clear, intuitive layout and labeled components	Mostly clear layout	Unclear or missing UI elements
Functionality	Fully sends and receives alerts via Bluetooth	Alerts sent but minor issues	App doesn't function as expected
Code Logic	All blocks organized and functional	Minor errors or block issues	Code is incomplete or missing logic
User-Centered Design	Designed with empathy, ease of use	Some accessibility elements	Lacks clarity or real-world focus
Testing & Reflection	Tested, improved, and shared reasoning	Basic test, minimal changes	No test or explanation of changes

Don't forget to go back and add this to improve part.





Now I Can

Objective: Students will reflect on their learning and identify how they've grown in understanding communication, empathy, and the role of assistive tools in helping others share ideas.

Activity Instructions:

1. Introduce the Reflection

- Say: "Let's look back on everything we've explored this lesson. You learned how it feels to face communication challenges, and how tools can help people share their thoughts in special ways."

2. Guide Students Through the Checklist

- Read each statement aloud one at a time.
- After each one, ask students to give a thumbs up, sideways, or down to show how confident they feel.
- Example prompts:
 - "Did you learn how people share ideas without talking?"
 - "Do you know how tools can help people talk?"

3. Student Completion

- Let students check off or color the "Now I Can" boxes they feel good about.
- Remind them: "It's okay if you didn't check all of them. We're all still learning."

4. Optional Extension

- Ask students to circle the one they're most proud of.
- Invite them to write a sentence or draw a picture to show their favorite learning moment.

Relation to Standards

1. NGSS (Next Generation Science Standards):

- **MS-ETS1-1:** Define a real-world problem related to health and safety and identify criteria for a successful solution — directly linked to designing an emergency alert system for elderly care.
- **MS-ETS1-2:** Evaluate competing solutions by testing different app layouts and Bluetooth alert functions — helping students determine which design best meets user needs.
- **MS-ETS1-3:** Develop and improve a working prototype based on feedback — aligned with iterative design and debugging in App Inventor.
- **MS-ETS1-4:** Use models to describe how communication happens between devices — demonstrated through interface layout and block-based logic flow.

2. ISTE Standards for Students (International Society for Technology in Education):

- **1. Empowered Learner:** Set goals and take ownership by using MIT App Inventor to solve a real-world problem in health care and safety.
- **4. Innovative Designer:** Apply design thinking to create and refine an assistive app that connects via Bluetooth and considers user experience (UX).
- **5. Computational Thinker:** Use logical sequences, input/output structures, and event-based programming to build interactive solutions.
- **6. Creative Communicator:** Present ideas and designs clearly through app prototypes and peer discussion.
- **7. Global Collaborator:** Work in teams to design for a global issue — supporting elderly users and improving safety through technology.

3. UN Sustainable Development Goals (SDGs):

- **SDG 3: Good Health and Well-Being:** Design an assistive app to promote safety, fast response, and well-being for the elderly.
- **SDG 4: Quality Education:** Engage in inclusive, real-world STEAM learning with access to technology and problem-solving.
- **SDG 9: Industry, Innovation & Infrastructure:** Explore how digital tools and innovation contribute to better health systems and user-centered product design.

4. CSTA (Computer Science Teachers Association) Standards:

- **Algorithms & Programming:** Use event-driven programming, sequencing, and logic blocks to simulate real-world behavior in the app.
- **Computing Systems:** Understand how devices (phones, sensors) communicate wirelessly using Bluetooth.
- **Data & Communication:** Develop an app that sends input from one device to another to simulate alert communication.
- **Impact of Computing:** Investigate how computing can enhance independence, health, and safety for specific users (e.g., elderly).

5. 21st Century Skills (P21 Framework):

- **Critical Thinking:** Solve coding and interface problems; debug and refine logic flows.
- **Communication:** Explain app functionality, UX choices, and how the system helps users.
- **Collaboration:** Partner with peers to design, test, and improve app features for real-world use.
- **Creativity:** Design custom layouts and alert messages that reflect empathy and user-centered thinking.

- **Digital Literacy:** Use real development platforms (MIT App Inventor) to create functional, purposeful technology.